VEDANT PAWAR

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PROFESSIONAL SUMMARY

Results-driven Python Developer and AI/ML Engineer with hands-on experience building production-grade NLP systems, LLM-integrated backends, and real-time AI applications, including Computer Vision and Object Detection using OpenCV, PIL/Pillow, YOLO, SSD, and Faster R-CNN. Architected CREVA — a 7-domain autonomous AI robot — using FastAPI, LangChain, RAG pipelines, and Docker, achieving 25% workflow efficiency gains across a 5-engineer team. Proficient in Python, Flask, Django, REST APIs, ChromaDB, OpenRouter, SerpAPI, ElevenLabs, AWS deployment patterns, CI/CD workflows, and Cloud Platforms. Actively seeking Python/AI backend or ML Engineer roles to deliver measurable business impact.

CORE TECHNICAL SKILLS

• Languages & Frameworks: Python · FastAPI · Flask · Django · REST API Development · SQL (MySQL · PostgreSQL) · JavaScript · HTML5 · CSS3

• AI / ML / NLP: LLM Integration · LangChain · RAG Pipelines · Prompt Engineering · NLP Pipelines · Hugging Face Transformers (basics) · AI Model APIs (OpenRouter

• SerpAPI) · Machine Learning · Data Preprocessing · Sentiment Analysis · Text Classification · Vector Databases (ChromaDB) · Computer Vision · Object Detection

• Voice & Multimodal AI: gTTS · ElevenLabs API · Multilingual Speech Generation · Voice-enabled Interaction

• Frontend & UI: Gradio · Streamlit · React (basics)

• Libraries & Data: Pandas · NumPy · Matplotlib · JSON · CSV Processing · SQLAlchemy (basics)

• DevOps

• Cloud & Tooling: Git · GitHub · Docker · Docker Compose · AWS (EC2

• S3 — deployment patterns) · CI/CD Pipelines · GitHub Actions (basics) · VS Code · Agile / Scrum · Cloud Platforms

• Core CS Concepts: OOP · Data Structures & Algorithms · API Design Patterns · Microservices Concepts · Scalable Architecture · Backend Optimization · Async Programming

PROFESSIONAL EXPERIENCE

AI/ML & IoT Intern — Incubators (CREVA AI Robot Project) Jan 2026 – Mar 2026

Team size: 5 engineers | Stack: Python · FastAPI · NLP · REST APIs · IoT integration

- Architected NLP-based interaction pipelines for the CREVA AI Robot, enabling real-time AI request processing across 7 functional domains and improving overall workflow efficiency by 25% — reducing average task completion time from ~40s to ~30s, utilizing Computer Vision techniques with OpenCV and PIL/Pillow.

- Integrated 5+ AI modules with web-based interfaces via RESTful APIs, achieving seamless cross-module communication; reduced inter-service API latency by ~30% through payload optimization and connection pooling.
- Designed and implemented a scalable AI + IoT backend architecture supporting modular, hot-swappable deployment workflows, enabling 3 new AI modules to be added without downtime, leveraging Docker and CI/CD Pipelines.
- Containerized the FastAPI backend using Docker, streamlining local development and enabling consistent, reproducible deployment environments across the 5-person engineering team.
- Collaborated in an Agile/Scrum environment — delivered across 6 two-week sprints, maintaining 100% on-time sprint delivery while iterating on real-time AI system integration.

KEY PROJECTS

CREVA AI System — Multi-Domain Autonomous AI Robot | Stack: Python · FastAPI · NLP · LangChain · Docker · REST APIs

• Designed and deployed NLP-driven interaction pipelines enabling real-time, context-aware AI communication across 7 independent functional domains — serving 20+ concurrent request types, incorporating Computer Vision with YOLO, SSD, and Faster R-CNN.

- Integrated multiple LLM and AI APIs into a unified, scalable FastAPI backend with LangChain orchestration, supporting hot-swappable modular AI workflows and reducing integration effort by ~40%.
- Engineered modular backend architecture with clean separation of concerns, containerized with Docker — enabling parallel feature development across a 5-engineer team with zero merge conflicts on core APIs.
- Implemented robust API communication with error handling, retry logic, and async request queuing, achieving 99%+ uptime during active testing across 7 functional domains.

EDUCATION

Bachelor of Computer Applications (BCA) 2022 – 2025

Shri Shivaji Science College, Amravati, Maharashtra

- Relevant coursework: Data Structures, Algorithms, Computer Networks, and Database Systems, providing a solid foundation for Computer Vision and AI/ML applications.

CERTIFICATIONS & CONTINUOUS LEARNING

• Python for Everybody — Coursera

- FastAPI & REST API Development — Self-directed (3 production projects)
- LangChain for LLM Application Development — DeepLearning.AI (In Progress)
- Machine Learning & NLP Fundamentals — Applied via CREVA AI and NLP System projects

- AWS Cloud Practitioner Essentials — In Progress (EC2, S3, Lambda deployment patterns applied in projects)
- Docker & Containerization — Applied across all 3 production GitHub repositories
- Git & GitHub Version Control — Active usage across 3+ production-grade repositories
- OpenCV and PIL/Pillow for Computer Vision — Applied in CREVA AI Robot Project and other personal projects.